Week 5 Introduction

A key truth of LLMs is that they are not always based on truth. They are based on so much information that there really is not "truth," just content put together based on sophisticated math. In addition, as you learned earlier, models have training cut-offs, and may be able to tell you everything there is to know about the human genome but can't tell you yesterday's weather in the town where you live. In addition, models can generate some very inappropriate content, which could be damaging to the reputations of the model makers (e.g., OpenAI, Google), so there had to be some kind of filtering mechanism for model responses.

Very early on it was clear more was required, and that's where retrieval augmented generation (RAG) comes in. Think of it conceptually as a filter that grounds the models in reality and enables prompt and response "editing." RAG is currently an important part of just about any large-scale use of generative AI.

This week has a **required** video titled Retrieval Augmented Generation with OpenAI/GPT and Chroma (https://www.youtube.com/watch?v=Cim1lNXvCzY), Learn Data with Mark, 2023. I'll have the setup information in my weekly video, and you'll need to understand this content to do this week's assignment.

Contents of the Week:

- Introduction
- Readings and Videos
- Week 5 Discussion/Participation
- Week 5 Exercise

Objectives/Topics:

- Understand what retrieval augmented generation is and how it's used.
- Describe the RAG architecture.
- Understand LangChain.

While there are only three objectives, they are each very deep, especially LangChain.

Helpful Resources:

- GitHub (https://github.com/) – A great resource to share the content you are creating with potential employers and connect with other data scientists in the community. We will be using this in our course, but it is also a great resource for you to continue using outside of your courses for any skunk work projects you are working on and for showing the world your expertise.
- Medium (https://medium.com/)– Aggregates articles on virtually any topic you can imagine, particularly data science. This is best to read daily.